

Dajin Lee, Ph. D. Candidate

Curriculum Vitae

Last Updated on May 10, 2026

Current Position and Contact Information

Position Ph. D. Candidate
Department of Convergence IT Engineering
Pohang University of Science and Technology (POSTECH)
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Research Interests

Haptic Perception & Rendering

- Spatial Tactile Rendering for Multiple Targets (CHI'26a)
- Discomfort from Steering Wheel Vibration (HS'26)
- Effects of Contact Force Across Upper Body (TOH'24a)
- Heart Activity Modulation Using Haptic Biofeedback (UIST'24)
- Body-Penetrating Phantom Sensation on Feet (WHC'21)

Multisensory Integration of Sound and Touch

- Perceptual Effects of Co-Located Vibrotactile and Auditory Cues (Submitted)
- Spatial Audio-Tactile Alignment for Directional Cueing (TVCG'25)
- Modulating Body-Penetrating Tactile Sensation Using Sound (WHC'25)

Human-Computer Interaction and Qualitative Analysis

- A Contextual Analysis of Social Touch (CHI'26b)
- Embodied and Tangible Learning for Children (IJHCS'25)
- Haptic Exhibit for Learning Physics (TOH'24b)
- Participatory Design for Vibrotactile Warning (Eurohaptics'24)

Education

2021.03 - Present **Ph. D. Candidate, Interaction Laboratory**
Computer Science and Engineering
Pohang University of Science and Technology (POSTECH)
Advisor: [Prof. Seungmoon Choi](#)

2013.03 - 2020.02 **Bachelor of Science**
Industrial Management and Engineering
Pohang University of Science and Technology (POSTECH)
Advisor: Prof. Chihyuk Jun

Publications

International Journal Articles

- [J1] **Dajin Lee**, Gyeore Yun, and Seungmoon Choi, "Effects of Contact Force on Vibrotactile Perceived Intensity Across the Upper Body," *IEEE Transactions on Haptics*, 2024 (Also presented in the IEEE Haptics Symposium 2024; **Candidate for the Best Paper Award**).
[Paper](#), [Presentation Slides](#)
- [J2] **Dajin Lee**, Daehyeon Nam, Jinhyuk Yoon, Dukchan Yoon, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi, "A Comparative Study of Physical and Haptic Exhibits in an Informal Learning Environment," *IEEE Transactions on Haptics*, 2024 (Also presented in the 2025 *IEEE World Haptics Conference*).
[Paper](#), [Demo Video](#)
- [J3] **Dajin Lee** and Seungmoon Choi, "Perceptual Alignment of Spatial Auditory and Tactile Stimuli for Effective Directional Cueing," *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025 (Also presented in the 2025 *IEEE Conference on Virtual Reality and 3D User Interfaces*).
[Paper](#), [Presentation Video](#)
- [J4] **Dajin Lee**, Daehyeon Nam, and Seungmoon Choi, "Augmenting Outdated Museum Exhibits with Embodied and Tangible Interactions for Prolonged Use and Learning Enhancement," *International Journal of Human-Computer Studies (IJHCS)*, 2025.
[Paper](#), [Demo Video](#)
- [J5] **Dajin Lee**, Uison Ju, and Seungmoon Choi, "Perceptual Effects of Co-Located Vibrotactile and Auditory Cues on the Head," *Springer VR*, 2026 (submitted).

International Conference Papers

- [C1] **Dajin Lee**, Seungjae Oh, Seungmoon Choi, and Bum-Jae You, "Vibrotactile Metaphor of Physical Interaction Using Body-Penetrating Phantom Sensations: Stepping on a Virtual Object," *IEEE World Haptics Conference (WHC)*, 2021.
[Paper](#), [Presentation Video](#)
- [C2] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi, "Participatory Design for In-Vehicle Vibrotactile Warnings on Driver's Seat," *EuroHaptics*, 2024.
[Paper](#), [Presentation Video](#), [Demo Video](#)
- [C3] Andreia Valente, **Dajin Lee**, Seungmoon Choi, Mark Billingham, and Augusto Esteves, "Modulating Heart Activity and Task Performance Using Haptic Heartbeat Feedback: A Study Across Four Body Placements," *ACM Symposium on User Interface Software and Technology (UIST)*, 2024.
[Paper](#), [Presentation Video](#)

- [C4] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi, "Enhancing Body-Penetrating Phantom Sensations Through Multisensory Integration of Sound and Vibration," *IEEE World Haptics Conference (WHC)*, 2025.
[Paper](#), [Supplementary Video](#)
- [C5] **Dajin Lee**, Soyeon Nam, Minseong Noh, and Seungmoon Choi, "Effects of Frequency and Magnitude on Steering Wheel Vibration Discomfort: A Hierarchical Modeling Approach to Individual Sensitivity," *IEEE Haptics Symposium*, 2026.
[Paper](#)
- [C6] Ayush Bhardwaj, Ashish Pratap, Abbas Khawaja, Yapeng Tian, Uison Ju, **Dajin Lee**, Seungmoon Choi, and Jin Ryong Kim, "Touch with Meaning: A Contextual Analysis of Social Touch," *ACM CHI conference on Human Factors in Computing Systems (CHI)*, 2026.
[Paper](#)

Domestic Conference Papers (Republic of Korea)

- [DC1] Dong Yeong Jeong, Sung H. Han, Seungmoon Choi, Mingyu Lee, Haewoo Kang, Sangyoon Han, Gyeore Yun, Hyo Seung Lee, Jiwan Lee, and **Dajin Lee**, "Motion and Haptic Effects Classification of a 4D Cinema Seat," *Ergonomics Society of Korea*, 2019.
- [DC2] Daehyeon Nam, **Dajin Lee**, and Seungmoon Choi, "Improving the Design of Science Museum Exhibits Through Digital Augmentation," *Korea Software Congress (KSC)*, 2022.
- [DC3] **Dajin Lee**, Gyeore Yun, and Seungmoon Choi, "Effects of Contact Force on Vibrotactile Perceived Intensity Across Upper Body," *Korea Haptics Conference (KHC)*, 2023.
- [DC4] **Dajin Lee** and Seungmoon Choi, "Differential Thresholds of Azimuth between Auditory and Tactile Stimuli," *Korea Haptics Conference (KHC)*, 2024 (**Best Paper Award**).

Posters & Demonstrations

- [P1] **Dajin Lee**, Daehyeon Nam, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi, "A Comparative Study of Physical and Haptic Exhibits in Informal Learning Environment," *IEEE World Haptics Conference (WHC)*, 2023 (Work in Progress).
- [P2] **Dajin Lee** and Seungmoon Choi, "Spatial Tactile Optimizer: Toward Perceptual Optimization for Multi-Point Vibrotactile Cues on the Torso," *ACM CHI conference on Human Factors in Computing Systems (CHI)*, 2026 (Extended Abstract).
[Paper](#)
- [D1] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi, "Participatory Design for In-Vehicle Vibrotactile Warnings on Driver's Seat," *EuroHaptics*, 2024.
[Paper](#), [Presentation Video](#), [Demo Video](#)
- [D2] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi, "Body-Penetrating Tactile Phantom Sensations Enhanced by Sound," *IEEE World Haptics Conference (WHC)*, 2025.
[Paper](#), [Supplementary Video](#)

Invited Talks and Tutorials

1. Seungmoon Choi and **Dajin Lee**, "Incorporating Qualitative Analysis in Haptics Research," *IEEE World Haptics Conference (WHC)*, 2025.
[Page](#), [Presentation Slides](#)

Honors & Awards

- 2026 **Special Recognition for Outstanding Review**
ACM SIGCHI 2026
- 2026 **Research Grant for Korea-Canada Research Training**
National Research Foundation of Korea (NRF)
- 2026 **POSTECH Metaverse Contents Contest (First Prize)**
Pohang University of Science and Technology (POSTECH)
- 2025 **ICT Challenge (Honorable Mention)**
Institute of Information & communications Technology Planning & Evaluation (IITP)
- 2024 **Best Paper Award**
Korea Haptics Conference (KHC)
- 2024 **Best Paper Award (Finalist)**
IEEE Haptics Symposium
- 2023 **Hyundai Student Innovation Challenge (Honorable Mention)**
Korea Haptics Conference (KHC)
- 2019 **POSTECH Hackathon (Popularity Award)**
Pohang University of Science and Technology (POSTECH)
- 2018 **POSTECH Hackathon (Excellence Award)**
Pohang University of Science and Technology (POSTECH)
- 2018 **Student Innovation Challenge (Excellence Award)**
Korean Institute of Industrial Engineers (KIIE)

Research Experiences

Research Internships

- 2017.12 - 2018.02 **Ship Routing Problem with Split Delivery and Time Windows**
Logistics Laboratory,
Industrial Management Engineering Department,
Pohang University of Science and Technology (POSTECH).
Advisor: Prof. Byung-In Kim
- 2019.01 - 2019.06 **Motion and Haptic Effects Classification of a 4D Cinema Seat**
2020.03 - 2020.08 *Interaction Laboratory,*
Computer Science and Engineering Department,

	Pohang University of Science and Technology (POSTECH). Advisor: Prof. Seungmoon Choi
2019.06 - 2019.08	3D Graphical User Interface in Virtual Reality Center for Intelligent & Interactive Robotics, Korea Institute of Science and Technology (KIST). Advisor: Dr. Jung-Min Park
2020.09 - 2020.12	EMG-based Control of Robotic Prostheses <i>Medical Assistant Robotics and Cognitive Haptics (MARCH) Laboratory,</i> Mechanical Engineering, Pohang University of Science and Technology (POSTECH). Advisor: Prof. Keehoon Kim
2025.07 - 2025.08	Development of a Taxonomy of Social Touch <i>Multimodal Interaction (MI) Laboratory,</i> Department of Computer Science, University of Texas at Dallas. Advisor: Prof. Jin Ryong Kim
2026.06 - 2026.11	Development of a Taxonomy of Social Touch <i>Haptic Experience (HX) Laboratory,</i> Department of Management Science and Engineering, University of Waterloo. Advisor: Prof. Oliver Schnieder

Research Projects

2019.01 - 2019.06	Automatic Authoring of Physical and Perceptual/Affective Motion Effects for Virtual Reality Samsung Research Funding & Incubation Center for Future Technology
2021.01 - 2022.12	Development and Validation of Real-Virtual Blended Exhibits for In-Depth Embodied Learning of Scientific Principles (Team Leader) Support Program for Reinforcement of Science Culture Exhibition Service, National Research Foundation (NRF)
2022.07 - 2023.06	Research on Function Advancement to Improve the Marketability of Vibration Seat (Team Leader) Hyundai Automobile Company
2022.09 - Present	Semantic Sound-to-Haptic Automatic Conversion: Metaverse, Full-body Haptic Effects, and Accessibility Mid-Career Researcher Program, National Research Foundation (NRF)
2025.01 - Present	Quantifying VOCs through Affective Natural Language Analysis of Steering Wheel Vibrations (Team Leader) Hyundai Automobile Company

Teaching Experiences

2019.09 - 2021.09	Introduction to Statistics for Data Science (Teaching Assistant) POSTECH & POSCO Group University
2024.09 - Present	Introductions to Haptics (Teaching Assistant) Pohang University of Science and Technology (POSTECH)

Academic Service

Reviewer	Eurohaptics (2024)
Reviewer	IEEE Transactions on Haptics (2025-2026)
Reviewer	IEEE Haptics Symposium (2026)
Reviewer	Frontiers in Psychology (2026)
Reviewer	ACM SIGCHI (2026)

Skills

Haptic Interaction	Tactile rendering/perception, Embodied learning, Authoring tool
Multisensory Perception	Spatial audio-tactile perception, Visuo-haptic illusion
Programming	C/C++/C#, MATLAB, Python, R
Statistics	Statistical analysis, Design of experiment